STA2002 Probability & Statistics II Lecture 2 - 2022-2023 Term 2

Lecture 2: Point Estimation

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Suggested Reading: Hogg et al. 2018, Chapter 6.

1 General Concepts of Point Estimation

The field of statistical inference consists of those methods used to make decisions or to draw conclusions about a **population**. These methods utilize the information contained in a **sample** from the population in drawing conclusions.

Suppose that we want to obtain a point estimate of a population parameter θ. We know that before the data is collected, the observations are considered to be random variables, say $X_1, X_2, \ldots, X_n$. Therefore, any function of the observation, or any statistic, is also a random variable. For example, the sample mean $\bar{X} = \frac{1}{n}\sum_i X_i$ and the sample variance $S^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2$ are statistics and they are also random variables. Since a statistic is a random variable, it has a probability distribution. We call the probability distribution of a statistic a **sampling distribution**.

We give the formal definition of estimator and estimate as follows.

Definition 1.1 (Estimator and Estimates)**.** *Suppose X is a random variable with pdf or pmf $f(x;\theta)$. Suppose we have a set of samples $X_1, \cdots, X_n$ from X, an estimator $\widehat{\theta} = \widehat{\theta}(X_1, \cdots, X_n)$ is a function of data (i.e., a formula), and an estimate is its value (i.e., a number).*

Example 1 Suppose we have a set of data $(X_1, X_2, X_3) = (0.5, 0.7, 1.0)$, then $\widehat{\theta} = \frac{1}{3}(X_1 + X_2 + X_3)$ is the estimator (formula), and $\widehat{\theta}_0 = \frac{1}{3}(0.5 + 0.7 + 1.0) = 0.733$ is the estimate (value).

Example 2 Suppose we toss coins whose probability of having head is p. In this case, p is the population parameter (independent of the realizations). If we want to estimate p, we independently toss the coins for 5 times, collecting samples $(X_1, \cdots, X_5)$, and derive the sample mean $\widehat{p} = \frac{1}{5}\sum_{i=1}^{5} X_i$. The formula we have is the estimator. Suppose one set of realizations is $(0, 1, 1, 0, 1)$, then the value $\widehat{p} = 0.6$ is the estimate.

Example 3 Suppose $X_1, \cdots, X_{10}$ are i.i.d. drawn from normal population $N(\mu, \sigma^2)$. In this case, the parameters μ and σ^2 are the population parameters. If we estimate by the sample mean $\frac{1}{10}\sum_{i=1}^{10} X_i$, then the sample mean will be the estimator.

Roughly speaking, as long as you can use data to calculate $\widehat{\theta}$'s value, then $\widehat{\theta}$ is an estimator. The population parameter is always a (unknown) given number independent of realizations, and an estimator always depends on data.

1.1 Bias, Variance, and MSE

We may have several different choices for the point estimator of a parameter. For example, if we wish to estimate the mean of a population, we might consider the sample mean or the sample median as point estimators. In order to decide which point estimator is the best one to use, we need to examine their statistical properties and develop some criteria for comparing estimators.

Now suppose we have a set of data (random variables) and an estimator $\widehat{\theta}(X_1, \cdots, X_n)$ for the underlying parameter θ. Note that $\widehat{\theta}(X_1, \cdots, X_n)$ is also a random variable since $X_1, \ldots, X_n$ are random. We thus define the bias, the variance, and the mean-square error (MSE) of $\widehat{\theta}(X_1, \cdots, X_n)$ as follows:

Definition 1.2 (Bias, Variance, Mean Square Error). *Suppose we have an estimator $\widehat{\theta} = \widehat{\theta}(X_1, \cdots, X_n)$ for the underlying parameter θ, then we define the bias, the variance, and the mean square error (MSE) as*

$$\begin{aligned} bias(\widehat{\theta}) &= \mathbf{E}[\widehat{\theta} - \theta] \\ \mathrm{Var}(\widehat{\theta}) &= \mathbf{E}[(\widehat{\theta} - \mathbf{E}\widehat{\theta})^2] \\ MSE(\widehat{\theta}) &= \mathbf{E}[(\widehat{\theta} - \theta)^2]. \end{aligned} \tag{1}$$

If $b = 0$, then we call $\widehat{\theta}$ the **unbiased** estimator.

Notice that

$$\begin{aligned} \mathrm{MSE}(\widehat{\theta}) &= \mathbf{E}(\widehat{\theta} - \mathbf{E}\widehat{\theta} + \mathbf{E}\widehat{\theta} - \theta)^2 \\ &= \mathbf{E}(\widehat{\theta} - \mathbf{E}\widehat{\theta})^2 + (\mathbf{E}\widehat{\theta} - \theta)^2 + 2\mathbf{E}(\widehat{\theta} - \mathbf{E}\widehat{\theta}) \times (\mathbf{E}\widehat{\theta} - \theta) \\ &= \{\mathrm{bias}(\widehat{\theta})\}^2 + \mathrm{Var}(\widehat{\theta}). \end{aligned}$$

Definition 1.3 (Consistency). *Suppose we have an estimator $\widehat{\theta} = \widehat{\theta}(X_1, \cdots, X_n)$, we say $\widehat{\theta}$ is consistent if for any given $\epsilon > 0$,*

$$\mathbf{P}(|\widehat{\theta} - \theta| > \epsilon) \to 0, \quad \text{as } n \to \infty, \tag{2}$$

here n is the sample size.

Example 4 Sample mean Suppose the observations are generated by i.i.d. random variables X_i. Let us consider sample mean $\widehat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i$ for the parameter $\theta = \mathbf{E}[X_1] = \mathbf{E}[X_i]$. By definition

$$\begin{aligned} \mathbf{E}[\widehat{\theta}] &= \frac{1}{n}\sum_{i=1}^n \mathbf{E}[X_i] = \theta \\ \mathrm{Var}(\widehat{\theta}) = \mathbf{E}(\widehat{\theta} - \mathbf{E}\widehat{\theta})^2 &= \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n \mathbf{E}(X_i - \theta)(X_j - \theta) = \frac{\mathrm{Var}(X_1)}{n} \end{aligned}$$

from Markov inequality, $\mathbf{P}\left(|\widehat{\theta} - \theta| > \epsilon\right) \le \dfrac{\mathbf{E}(\widehat{\theta} - \theta)^2}{\epsilon^2} = \dfrac{\mathrm{Var}(X_1)}{n\epsilon^2} \to 0, \text{ as } n \to \infty$

thus the sample mean is **unbiased** and **consistent** estimator for the true mean value $\theta = \mathbf{E}[X_1]$.

Example 5 Sample variance Suppose the observations are generated by i.i.d. random variables X_i. Let us consider sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$ for the parameter $\sigma^2 = \mathrm{Var}[X_1]$. By definition

$$\mathbf{E}[S^2] = \frac{1}{n-1}\mathbf{E}[\sum_{i=1}^n X_i^2 - n\bar{X}^2] = \frac{1}{n-1}[n\{\mathbf{E}[X_1]^2 + \sigma^2\} - n\{\mathbf{E}[X_1]^2 + \frac{\sigma^2}{n}\}] = \sigma^2,$$

thus the sample variance is **unbiased** estimator for the true variance value $\mathrm{Var}(X_1)$.

Example 6 Now suppose $X_1, \cdots, X_n$ satisfy Bernoulli distribution $B(1,p)$, here p is the probability of getting 1. For $\mathbf{E}[X_1] = 1 \times p + 0 \times (1-p) = p$, so the sample mean $\widehat{p} = \frac{1}{n}\sum_{i=1}^n X_i$ is a good estimator for p.

Now suppose we have an extra information that $p \in [\frac{1}{2}, 1]$, consider the new estimator

$$\widehat{p}^* = \begin{cases} \widehat{p}, \text{ if } \widehat{p} \geq 1/2, \\ \frac{1}{2}, \text{ if } \widehat{p} < 1/2 \end{cases}$$

Since

$$\mathbf{E}(\widehat{p}^* - p)^2 = \mathbf{E}[(\widehat{p} - p)^2 \times \mathbf{1}_{\widehat{p} \geq 1/2} + (\frac{1}{2} - p)^2 \mathbf{1}_{\widehat{p} < 1/2}] = \mathbf{E}[(\widehat{p} - p)^2 \times \mathbf{1}_{\widehat{p} \geq 1/2}] + (\frac{1}{2} - p)^2 \times \mathbf{P}(\widehat{p} < 1/2),$$

we have

$$\mathbf{E}(\widehat{p} - p)^2 - \mathbf{E}(\widehat{p}^* - p)^2 = \mathbf{E}[(\widehat{p} - p)^2 \times \mathbf{1}_{\widehat{p} < 1/2}] - (\frac{1}{2} - p)^2 \times \mathbf{P}(\widehat{p} < 1/2) \geq 0. \text{ (because } p \geq 1/2)$$

Therefore we have $\mathrm{MSE}(\widehat{p}^*) \leq \mathrm{MSE}(\widehat{p})$ when $p \in [\frac{1}{2}, 1]$.

Example 7 Suppose $X_1, X_2, \cdots, X_n$ are i.i.d. Poisson random variables with parameter λ. In this case, we define the estimator

$$\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^n X_i.$$

Since

$$\mathbf{E}\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^n \mathbf{E}X_i = \lambda,$$

the estimator is unbiased. On the other hand,

$$Var(\widehat{\lambda}) = \frac{1}{n^2}\sum_{i=1}^n Var(X_i) = \frac{\lambda}{n},$$

so the MSE of $\widehat{\lambda}$ is $\frac{\lambda}{n}$.

2 Method of Moment (MoM)

Suppose we want to estimate a parameter θ, and fortunately it can be written as an expectation (population moment), i.e., $\theta = \mathbf{E}[h(X_1)]$, then we can replace '$\mathbf{E}[h(X_1)]$' by $\frac{1}{n}\sum_{i=1}^n h(X_i)$ to derive our estimator. **Remember to replace all of the expectation (i.e., population moment) by average (i.e., sample moment).**

Definition 2.1. *Let $X_1, \ldots, X_n$ be a random sample. The k-th population moment is $\mathbf{E}[X^k]$, $k = 1, 2, \ldots$ The corresponding k-th sample moment is $\frac{1}{n}\sum_{i=1}^n X_i^k$, $k = 1, 2, \ldots$.*

For example, denote the first population moment as the population parameter $\mu = \mathbf{E}[X]$ and the first sample moment is $\frac{1}{n}\sum_i X_i = \bar{X}$. By equating the population and the sample moment, we obtain the estimate for μ as $\hat{\mu} = \bar{X}$, which we call the method of moment estimator.

In general, suppose we have multiple unknown parameters $\theta_1, \ldots, \theta_m$, we will need to equate the first m population and sample moments to derive the method of moment estimator.

Definition 2.2 (Method of Moments). *Let $X_1, \ldots, X_n$ be a random sample from a probability distribution with unknown parameters $\theta_1, \ldots, \theta_m$. The moment estimators $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.*

The underlying idea of MoM comes from the law of large number. Notably, MoM is a method rather than a theorem, so you still need to verify your estimator after deriving it.

Example 8 Find MoM estimator for variance Suppose the data are i.i.d. By definition

$$\mathrm{Var}(X_1) = \mathbf{E}(X_1 - \mathbf{E}[X_1])^2 = \mathbf{E}[X_1^2] - (\mathbf{E}[X_1])^2$$

Replace $\mathbf{E}[X_1^2]$ by $\frac{1}{n}\sum_{i=1}^n X_i^2$, and replace $\mathbf{E}[X_1]$ by $\frac{1}{n}\sum_{i=1}^n X_i$, then we get the moment estimator for $\mathrm{Var}(X_1)$:

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \frac{1}{n}\sum_{i=1}^n X_i)^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2.$$

From this example, we know that given $X_1, \ldots, X_n$ i.i.d. sampled from the normal distribution $N(\mu, \sigma^2)$, the moment estimator for the mean is $\hat{\mu} = \bar{X}$ and for variance is $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$.

Example 9 Find MoM estimator for uniform distribution Suppose the data $X_1, \ldots, X_n$ are i.i.d. sampled from the uniform distribution on $[0, \theta]$. Then the population moment is $\mathbf{E}[X_1] = \frac{\theta}{2}$. Equating $\frac{\theta}{2} = \bar{X}$, thus the moment estimator is $\hat{\theta} = 2\bar{X}$.

Example 10 Suppose $X_1, \cdots, X_n$ are i.i.d. Poisson λ distributed random variables. Since $\lambda = \mathbf{E}X_1$, the MoM estimator for λ can be

$$\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Alternatively, for $Var(X_1) = \lambda$, in this case, we can also estimate λ by

$$\widehat{\lambda}^{\dagger} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \frac{1}{n}\sum_{i=1}^{n} X_i)^2.$$

Now, suppose we want to estimate the skewness of X_1. Since the skewness of Poisson distribution is $1/\sqrt{\lambda}$, the MoM estimator for the skewness is $1/\sqrt{\widehat{\lambda}}$.

Example 11 Suppose $X_1, \cdots, X_n$ are i.i.d. random variables. For any set $A \subset \mathbf{R}$, notice that

$$Prob(X_1 \in A) = \mathbf{E}1_A(X_1),$$

so the MoM estimator for $Prob(X_1 \in A)$ is

$$\widehat{q} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_A(X_i).$$

Algorithms to compute MoM estimates Solving a system of non-linear equations analytically is typically challenging and tedious. With the help of numerical solvers, there are basically two general strategies to compute MoM estimates.

The first strategy is to use the numerical solvers of nonlinear equations. In R, one such function is `nleqslv`. We showcase the use of the function in the following example. **Example 12 Uniform** Suppose $X_1, ..., X_n \overset{iid}{\sim} Unif(\theta_1, \theta_2)$. We use the first two moments to construct the estimator:

$$\begin{aligned} \tfrac{1}{n}\textstyle\sum_{i=1}^{n} X_i = &\quad \tfrac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) \\ \tfrac{1}{n}\textstyle\sum_{i=1}^{n} X_i^2 = &\quad \tfrac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1\hat{\theta}_2 + \hat{\theta}_2^2). \end{aligned}$$

The moment matching errors in presented in Figure 1:

$$\begin{aligned} \text{the first moment's matching error } &= \quad \tfrac{1}{2}(\theta_1 + \theta_2) - \tfrac{1}{n}\textstyle\sum_{i=1}^{n} X_i \\ \text{the second moment's matching error } &= \quad \tfrac{1}{3}(\theta_1^2 + \theta_1\theta_2 + \theta_2^2) - \tfrac{1}{n}\textstyle\sum_{i=1}^{n} X_i^2. \end{aligned}$$

The intersection of the two surfaces and the horizontal plane is the MoM estimate.

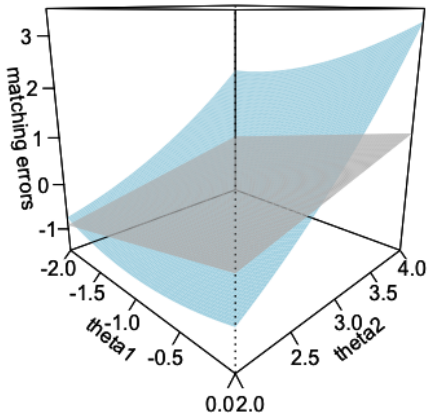


Figure 1: Surfaces of moment matching errors.

```
X <- runif(100, min = -1, max = 3)
angle = 45
persp(seq(-2,0,length.out = 100),
      seq(2,4, length.out = 100),
      outer(seq(-2,0,length.out = 100),
            seq(2,4, length.out = 100),
            function(x,y) {(x^2+y^2+x*y)/3 - mean(X^2)}),
      xlab = "theta1", ylab = "theta2", zlab = "matching errors",
      zlim = c(-1.5,3.5),
      col='lightblue',ticktype = "detailed",border = NA,theta = angle)
par(new=TRUE)
persp(seq(-2,0,length.out = 100),
      seq(2,4, length.out = 100),
      outer(seq(-2,0,length.out = 100),
            seq(2,4, length.out = 100),
            function(x,y) {(x+y)/2 - mean(X)}),
      xlab = "theta1", ylab = "theta2", zlab = "matching errors",
      zlim = c(-1.5,3.5),
      col='grey',ticktype = "detailed",border = NA,theta = angle)
```

We use `nleqslv` to solve the following equations:

$$\begin{aligned} \frac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) - \frac{1}{n}\sum_{i=1}^{n} X_i &= 0 \\ \frac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1\hat{\theta}_2 + \hat{\theta}_2^2) - \frac{1}{n}\sum_{i=1}^{n} X_i^2 &= 0. \end{aligned}$$

```
moments.eq <- function(theta){
```

```
    m1 <- (theta[1]+theta[2])/2
    m2 <- (theta[1]^2+theta[2]^2+theta[1]*theta[2])/3
    return(c(m1-mean(X),m2-mean(X^2)))
}
nleqslv(c(min(X)-10,max(X)+10), moments.eq)
```

The second strategy is to turn the problem of equation solving to optimization. Instead of solving the moment equations directly, we choose the parameters to minimize the ℓ_2 loss of the moment matching errors.

Suppose we have p moment conditions, the optimization problem is to minimize

$$\sum_{k=1}^{p} \left(\frac{1}{n} \sum_{i=1}^{n} X_i^k - m_k(\theta) \right)^2$$

with respect to $\theta \in \Theta$. Since we have p parameters, the optimal value of the objective function should be 0.

In this case, we use optimization algorithms to solve for the MoM estimates. In R, the function `optim` is a general purpose optimization function.

Example 13 Uniform, continued Under the same setup, the objective function to be minimized is

$$\left(\frac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) - \frac{1}{n} \sum_{i=1}^{n} X_i \right)^2 + \left(\frac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1 \hat{\theta}_2 + \hat{\theta}_2^2) - \frac{1}{n} \sum_{i=1}^{n} X_i^2 \right)^2 .$$

Next we use `optim` to numerically solve for the MoM estimates.

```
moments.opt <- function(theta, X){
    m1 <- (theta[1]+theta[2])/2
    m2 <- (theta[1]^2+theta[2]^2+theta[1]*theta[2])/3
    return((m1-mean(X))^2+(m2-mean(X^2))^2)
}
X <- runif(100, min = -3, max = 5)
optim(c(min(X)-1,max(X)+1), moments.opt,X=X)
```

2.0.1 Asymptotic theory: Consistency

Section 2 describes a few algorithms to produce the point estimates of the method of moments. In the simulation examples, the point estimates are very close to the true values. One may expect that with more and more observations, the method of moments estimates will be more and more accurate. Indeed, the method of moments estimators are consistent if the solution of the moment conditions is continuous.

For many cases, consistency of the method of moments estimators is immediate by law of large numbers. For instance, the mean parameter of Bernoulli random variables, Gaussian random variables, and many others.

Example 14 Bernoulli mean Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_0)$. The MoM estimator for θ_0 is $\hat{\theta}_{M0M} := \frac{1}{n}\sum_{I=1}^{n} X_i$. By LLN, $\hat{\theta}_{MoM} \xrightarrow{p} \theta_0$.

For more complicated problems, the method of moments estimators are not directly sample averages (e.g. the toy example of inferring the endpoints of a univariate uniform distribution), so LLN cannot be applied directly to show consistency. In general, consistency of MoM estimators is established by both the law of large numbers and the *continuous mapping theorem* (CMT).

The continuous mapping theorem states that the convergence in probability and distribution is preserved under continuous transformations.

Theorem 2.1 (Continuous Mapping Theorem (CMT))**.** *Let $X_n, n = 1, ...,$ and X be random variables. Let $g(\cdot)$ be a continuous function. If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*

Though Theorem 2.1 is stated for random variables, the same type of results also hold for random vectors and multivariate continuous functions. In the case of random vectors, $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ if their Euclidean distance $||\mathbf{X}_n - \mathbf{X}||$ (or in ℓ_2 norm) converges to 0 in probability.

Convergence results are preserved under arithmetic operations. As arithmetic operations are continuous, such results are also known as continuous mapping theorem. The following can be implied by the random vector version of the CMT.

Theorem 2.2 (CMT for arithmetic operations)**.** *Let X_n, X, Y_n, Y be random variables, and c be some constant. Then*

- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$.*
- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n Y_n \xrightarrow{p} XY$.*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n + Y_n \xrightarrow{d} X + c$. (Slutzky's theorem)*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n Y_n \xrightarrow{d} cX$. (Slutzky's theorem)*

Now with the CMT, we can prove MoM estimators are consistent under mild conditions.

Theorem 2.3 (Consistency of MoM estimators)**.** *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x; \theta_0)$, the $p-$dimensional parameter $\theta_0 \in \Theta \subseteq \mathbf{R}^p$, and the first p moments exist. That is, in the moment conditions*

$$\begin{aligned} \mathbf{E}_\theta[X_1] &= m_1(\theta) \\ \mathbf{E}_\theta[X_1^2] &= m_2(\theta) \\ &... \\ \mathbf{E}_\theta[X_1^p] &= m_p(\theta), \end{aligned}$$

the functions $\{m_j(\theta)\}_{j=1}^p$ are all well-defined. Assume there exists a unique solution to the system of moment equations:

$$\theta = g\left(\mathbf{E}_\theta[X_1], \mathbf{E}_\theta[X_1^2], ..., \mathbf{E}_\theta[X_1^p]\right),$$

and $g(\cdot)$ is continuous. Then, the method of moments estimator

$$\hat{\theta} = g\left(\frac{1}{n}\sum_{i=1}^n X_i, \frac{1}{n}\sum_{i=1}^n X_i^2, ..., \frac{1}{n}\sum_{i=1}^n X_i^p\right) \xrightarrow{p} \theta_0.$$

Proof By LLN, $\frac{1}{n}\sum_{i=1}^n X_i^k \xrightarrow{p} \mathbf{E}_{\theta_0}[X_1^k]$ for all $k = 1, ..., p$. In fact, the convergence in probability for all p moments hold simultaneously. (A simple application of the union bound.)

By continuous mapping theorem and the assumption that $g(\cdot)$ is continuous, consistency of the method of moments estimator follows. □

The key ingredient is the assumption that $g(\cdot)$ is continuous. For many parametric models, the moments depend on the parameter values continuously, that is, the moments are continuous in the parameters. As the number of equations is the same as the number of unknowns/parameters, the inverse function $g(\cdot)$ can also be continuous under very mild conditions (Brouwer invariance of domain theorem). Indeed, the continuity assumption is a mild assumption. In real data applications, this assumption is verifiable.

2.0.2 The "plug-in" estimator

In general, practitioners are not always intersted in the model parameters. The parameter of interest, say η, is often a *continuous* function of the model parameters θ, say $\eta = g(\theta)$.

Example 15 Effect size To evaluate the effect size of a vaccine, the treatment group is assigned the vaccine and the outcomes of the individuals of the treatment group are denoted as $\{X_1, ..., X_n\}$, and the control group is assigned the placebo and the outcomes are denoted as $\{Y_1, ..., Y_m\}$. Since the assignment of the treatment status and the control status is random, it is natural to assume $\{X_i\}$ and $\{Y_j\}$ are independent, and within each group, the outcomes are iid.

Suppose the data is the status of the infection with 1 denoting infected and 0 denoting not infected. (In other applications, the outcomes could be some other metrics, e.g., blood pressure.) Therefore, the data is modelled as two independent groups of Bernoulli random variables $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_1)$, and $Y_1, ..., Y_m \overset{iid}{\sim} Ber(\theta_2)$ for some true parameters $(\theta_{0,1}, \theta_{0,2})$. The parameter of interest is the effect size:

$$\eta = g(\theta_1, \theta_2) = \theta_1 - \theta_2.$$

Suppose we have a consistent estimator $\hat{\theta}$ for θ, we can *plug in* the estimator $\hat{\theta}$ to the continuous function $g(\cdot)$, and then the estimator for η,

$$\hat{\eta} := g(\hat{\theta}),$$

is a consistent estimator for η by continuous mapping theorem.

Example 16 Effect size, continued The plug-in estimator for $\eta \equiv \theta_1 - \theta_2$ is

$$\hat{\eta} = g(\hat{\theta}_1, \hat{\theta}_2) = \hat{\theta}_1 - \hat{\theta}_2.$$

As there is seldom lack of consistent estimators (the method of moments estimators are often consistent) and the parameter(s) of interest is often a continuous function of model parameters, the idea of "plug-in" is widely applicable.

Example 17 Bernoulli variance Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_0)$, and our main interest is in the variance of X_1:

$$V(\theta_0) \equiv \theta_0(1 - \theta_0).$$

If we write down the moment conditions for both the mean and the variance directly, then we would end up with two equations but only one unknown. In general, there is no solution to the system of equations with more unknowns than equations. (unless some equations are redundant.)

Since the variance of a Bernoulli random variable is a continuous function of the mean parameter, we can construct a consistent estimator for the variance using the method of moments estimator of the mean parameter. That is, a consistent estimator of the variance can be

$$\hat{V}_{plug-in} := \bar{X}_n(1 - \bar{X}_n)$$

where $\bar{X}_n = \frac{1}{n}\sum_{I=1}^{n} X_i$.

By LLN and continuous mapping theorem, $\hat{V}_{plug-in} \xrightarrow{p} V(\theta_0)$.

3 Maximum Likelihood Estimation (MLE)

In the maximum likelihood estimation, we assume that the data vector $(x_1, \cdots, x_n)$ are generated by a parametric distribution, i.e., the joint probability mass / probability density function is parameterized by some parameter $\theta \in \mathbf{R}^p$:

$$p(x_1 = i_1, \cdots, x_n = i_n) = p(x_1 = i_1, \cdots, x_n = i_n; \theta) \qquad \text{discrete distribution; pmf}$$

$$f(x_1, \cdots, x_n) = f(x_1, \cdots, x_n; \theta) \qquad \text{continuous distribution; pdf.}$$

Once we get the observed sample values $x_1, \ldots, x_n$, define the **likelihood function**:

$$L(\theta) = \begin{cases} p(x_1 = i_1, \cdots, x_n = i_n; \theta), & \text{for discrete distribution} \\ f(x_1, \cdots, x_n; \theta), & \text{for continuous distribution} \end{cases}$$

For example, suppose X is a random variable with probability distribution $f(x; \theta)$. Let $x_1, \ldots, x_n$ be the observed values in an i.i.d. random sample of size n, then the likelihood function of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots f(x_n; \theta).$$

Note that the likelihood function is now a function of only the unknown parameter θ, since $x_1, \ldots, x_n$ are the observed sample values.

Example 18 Suppose the data vector is $(x_1, x_2, \cdots, x_7) = (1, 1, 0, 0, 1, 0, 1)$. To model the data, we can assume that the data are generated by i.i.d. Binomial distribution with parameter p, i.e., $\mathbf{P}(x_i = t) = p^t(1-p)^{1-t}$, $t = 0, 1$. Then the joint probability mass function will be

$$\begin{aligned} \mathbf{P}\left(x_1 = t_1, \cdots, x_7 = t_7\right) &= p^{t_1}(1-p)^{1-t_1} \times \cdots \times p^{t_7}(1-p)^{1-t_7} \\ &= p^{\sum_{i=1}^7 t_i}(1-p)^{7-\sum_{i=1}^7 t_i}. \end{aligned}$$

By plugging in $(x_1, \cdots, x_7) = (1, 1, 0, 0, 1, 0, 1)$, we have the likelihood function (as a function of the parameter p) is:

$$L(p) = p^4(1-p)^3.$$

Example 19 Suppose the data vector $(x_1, x_2, \cdots, x_n)$. To model this data, we assume that the data are generated by normal random variables with different mean μ_i and a common variance σ^2, i.e.,

$X_i \sim N(\mu_i, \sigma^2)$. Then the joint density of $X_1, \cdots, X_n$ will be

$$f_{X_1,\cdots,X_n}(t_1, \cdots, t_n) = \frac{1}{(2\pi\sigma^2)^{n/2}} \prod_{i=1}^{n} \exp\left(-\frac{(t_i - \mu_i)^2}{2\sigma^2}\right). \tag{3}$$

Case 1. Suppose $\mu_1 = \mu_2 = \cdots = \mu_n = \mu$ and σ^2 is known, then once we gather the data $(x_1, \cdots, x_n)$ (some numbers), the likelihood function becomes

$$L(\mu) = \frac{1}{(2\pi)^{n/2}\sigma^n} \prod_{i=1}^{n} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right). \tag{4}$$

Case 2. Suppose $\mu_i = \mu$, but σ^2 is not known, then the likelihood function, as a function of both μ and σ^2, becomes

$$L(\mu, \sigma^2) = \frac{1}{(2\pi)^{n/2}\sigma^n} \prod_{i=1}^{n} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right). \tag{5}$$

Case 3. Suppose $\mu_1, \cdots, \mu_n$ are unknown and σ^2 is known, then the likelihood function becomes

$$L(\mu_1, \cdots, \mu_n, \sigma^2) = \frac{1}{(2\pi)^{n/2}\sigma^n} \prod_{i=1}^{n} \exp\left(-\frac{(x_i - \mu_i)^2}{2\sigma^2}\right). \tag{6}$$

i.e., even if the data set is the same, once you change the model, the likelihood function will be different. Also, you may use different models to model the same data set, which leads to the model selection problem.

Example 20 Suppose $X_1, \cdots, X_n$ are i.i.d. Poisson random variables, whose probability mass function is

$$Prob(X_1 = k) = \frac{\lambda^k}{k!} \exp(-\lambda).$$

In that case, the likelihood function for λ is

$$L(\lambda) = \lambda^{\sum_{i=1}^{n} X_i} \exp(-n\lambda)/(X_1! X_2! \cdots X_n!).$$

Definition 3.1 (**Maximum likelihood estimator).** *Once we have the likelihood function, we define the maximum likelihood estimator as*

$$\widehat{\theta} = \operatorname*{argmax}_{\theta} L(\theta), \text{ which is equivalent as } \widehat{\theta} = \operatorname*{argmax}_{\theta} \ell(\theta) := \log L(\theta))$$

where $\ell(\theta) := \log L(\theta))$ is also called the log-likelihood function.

Note that in many cases it is much easier to find the MLE through maximizing the log-likelihood function.

Example 21 Binomial distribution In example 18, we derive the likelihood function of binomial distribution

$$L(p) = p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}, \text{ and } \ell(p) = \log(L(p)) = (\sum_{i=1}^n x_i)\log(p) + (n - \sum_{i=1}^n x_i)\log(1-p)$$

Setting the derivative to 0, we have

$$\ell'(p) = \frac{\sum_{i=1}^n x_i}{p} - \frac{n - \sum_{i=1}^n x_i}{1-p} = 0 \Rightarrow \hat{p} = \frac{1}{n}\sum_{i=1}^n x_i = \bar{x}.$$

Moreover, in order to make sure $\hat{p}$ is indeed the maximizer, we also need to check the second-order derivative:

$$\ell''(\hat{p}) = -\frac{\sum_{i=1}^n x_i}{\hat{p}^2} - \frac{n - \sum_{i=1}^n x_i}{(1-\hat{p})^2} < 0,$$

Therefore, we have $\ell'(\hat{p}) = 0$ and $\ell''(\hat{p}) \leq 0$, which means that $\hat{p}$ is indeed a local maximizer of $\ell(p)$, thus $\hat{p} = \bar{x}$ is the MLE. (Do not forget to verifying the second-order derivative!)

Notably, sometimes you cannot find the maximizer via setting the derivative to 0, in that case you need to use the definition to find the maximum likelihood estimator.

Example 22 Uniform distribution Suppose the data samples $x_1, \ldots, x_n$ are sampled from uniform distribution on $[0, \theta]$ with unknown parameter θ. The likelihood function thus is defined as

$$L(\theta) = \frac{1}{\theta}\mathbf{1}\{0 \leq x_1 \leq \theta\} \times \cdots \times \frac{1}{\theta}\mathbf{1}\{0 \leq x_n \leq \theta\} = \frac{1}{\theta^n}\mathbf{1}\{0 \leq \min\{x_1, \ldots, x_n\}, \max\{x_1, \ldots, x_n\} \leq \theta\}.$$

Thus the likelihood function equals $\frac{1}{\theta^n}$ when $\theta \geq \max\{x_1, \ldots, x_n\}$ and equals 0 when $\theta < \max\{x_1, \ldots, x_n\}$. Therefore, the likelihood function is maximized when $\theta = \max\{x_1, \ldots, x_n\}$ and we conclude $\max\{x_1, \ldots, x_n\}$ is the maximum likelihood estimator of θ.

Example 23 Continue of example 19 Now we consider 3 cases in example 19. In case 1, the log likelihood function will be

$$\ell(\mu) = -\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} - \frac{n}{2}\log(2\pi\sigma^2) \tag{7}$$

Taking derivative and set it to 0, we have

$$\ell'(\mu) = \frac{1}{\sigma^2}\sum_{i=1}^n (x_i - \mu) = 0 \Rightarrow \hat{\mu} = \frac{1}{n}\sum_{i=1}^n x_i = \bar{x}.$$

(Please verify the second-order derivatives.)

In case 2, the variance is also not known, so we solve the following two equation together:

$$\begin{cases} \frac{\partial \ell(\mu,\sigma^2)}{\partial \mu} = 0 \\ \frac{\partial \ell(\mu,\sigma^2)}{\partial \sigma^2} = \sum_{i=1}^n \frac{(x_i-\mu)^2}{2\sigma^4} - \frac{n}{2}\frac{1}{\sigma^2} = 0 \end{cases} \Longrightarrow \begin{cases} \hat{\mu} = \frac{1}{n}\sum_{i=1}^n x_i \\ \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (x_i - \hat{\mu})^2 \end{cases} \tag{8}$$

In case 3, each μ_i is different, so the maximizer becomes

$$\frac{\partial \ell(\mu_1, \cdots, \mu_n, \sigma^2)}{\partial \mu_i} = 0 \Rightarrow x_i - \mu_i = 0 \Rightarrow \hat{\mu}_i = x_i$$

which estimates μ_i only via one sample x_i.

Example 24 In Example 20, the log–likelihood function is

$$\ell(\lambda) = \sum_{i=1}^n X_i \log(\lambda) - n\lambda - \log(X_1! \cdots X_n!), \tag{9}$$

so

$$\ell'(\lambda) = 0 \Rightarrow \frac{1}{\lambda}\sum_{i=1}^n X_i - n = 0 \Rightarrow \widehat{\lambda} = \frac{1}{n}\sum_{i=1}^n X_i.$$

3.1 Equivariance of Maximum likelihood estimators

By inspecting the optimization problem of MLE, it is immediate that MLE is equivariant. This is in the same idea as "plug-in".

Theorem 3.1 (Equivariance of MLE)**.** *Let $\tau = g(\theta)$ be a function of θ. Let $\hat{\theta}$ be the MLE of θ. Then, $\hat{\tau} = g(\hat{\theta})$ is the MLE of τ.*

Proof omitted. □

Example 25 Suppose the concentration of some chemical decays exponentially:

$$\mathbf{E}[Y(t)] = Y(0)e^{-\lambda t}$$

where t denotes time and λ is some rate parameter that controls the speed of the decay. The chemists are interested in the half life time which is the time point where the concentration is half of its initial value. By direct calculation, half time is $\frac{\log(2)}{\lambda}$. Then, its MLE is $\frac{\log(2)}{\hat{\lambda}}$ where $\hat{\lambda}$ is the MLE of λ.

Example 26 Let $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$. The MLE for θ is $\hat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i$.

- The MLE for $\tau = e^\theta$ is $\hat{\tau} = e^{\frac{1}{n}\sum_{i=1}^n X_i}$.
- The MLE for $\tau = \theta^2$ is $\hat{\tau} = \left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2$.
- The MLE for $\tau = \mathbb{1}(\theta \geq 0)$ is $\hat{\tau} = 1\left(\frac{1}{n}\sum_{i=1}^n X_i \geq 0\right)$.

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*